ANALYTICS REPORT

TO: E AND J GALLO WINERY

FROM: CHRIS WILLIAMS

SUBJECT: CASE ASSIGNMENT #1: WINE DATA PRELIMINARY ANALYSIS

DATE: 07/18/2024

Introduction

This report analyzes data on the prices and ratings of Red, White, Sparkling, and Rose wines to explore their relationships and differences. Our findings show that Sparkling wine has the highest average price and rating, while Rose wine has the lowest in both categories. Ratings were consistent across wine types, but prices varied significantly. We recommend Gallo focus on moderately priced, highly rated wines and target premium customers with Sparkling wine, while reconsidering the lower-rated, lower-priced Rose wine for better profitability and customer satisfaction.

Data Analysis

Discussion of Descriptive Statistics

The pair of wines that had the largest difference in sample average wine price was was sparkling wine and rose wine with an estimated difference of 48. The pair with the smallest difference was red and white with an estimated difference of 19.

Red Wine:
The ratings are very close to symmetric with a very slight left skew (-0.06), while the price is skewed to the right (13.40).

White Wine:
The ratings are symmetrical (-0.05), while the price is skewed to the right (9.72).

Sparkling Wine:
The ratings is symmetrical with a slight skew to the right (0.12), while price is also symmetrical with a slight skew to the right (2.70).

Rose Wine:
The rating is symmetrical like sparkling (0.09), while price is very much right skewed (9.58).

All around for ratings, it does not seem like there may be any notable outliers, while price will have some outliers with the range of prices for each wine varying from low prices to extremely high prices. For example, red wine has a minimum of $3.55 and a maximum of $3410.79.

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Discussion of ANOVA

Rating:

H_0: The average wine rating is equal across the 4 wine types (Red, White, Sparkling, Rose)

H_A: There is at least one difference in average wine rating across the 4 wine types (Red, White, Sparkling, Rose)

Because the p-value of 2.35E-74 is less than our significance level of 0.05, we can reject the null hypothesis and can conclude that there is at least one difference in average wine rating across the 4 wine types

At least one recommendation to Gallo based on this finding and what we see in the sample averages is to start selling more sparkling wine due to that being the highest rated wine out of the 4 wine types (Red, White, Sparkling, Rose).

Price:

H_0: The average wine price is equal across the 4 wine types (Red, White, Sparkling, Rose)

H_A: There is at least one difference in average wine price across the 4 wine types (Red, White, Sparkling, Rose)

Because the p-value of 1.55E-52 is less than our significance level of 0.05, we can reject the null hypothesis and can conclude that there is at least one difference in average wine price across the 4 wine types

At least one recommendation to Gallo based on this finding and what we see in the sample averages is to raise the average price of white wine by a little because red and white are very similarly rated, and red wine has a higher average than white.

Discussion of Bar Graph

The relationship between price and rating is directly correlated in highest rated and highest price, and lowest rated and lowest price. Sparkling has the highest price and the highest rating which means there's a direct correlation with it being the more expensive wine it is the higher rated. With the lowest priced wine being rose and rose also being the lowest rated, there's is a clear correlation between price and rating.

The pricing is not consistent as each wine all have signific differences in average price while rating seems to be very close between all 4 wine types.

Gallo can use this information to understand that high prices do not necessarily correlate with high ratings. This insight can guide them to focus on offering wines that provide good value—moderately priced wines with higher ratings. Additionally, knowing that Sparkling wine has the highest price and a high rating, they could target marketing efforts towards premium customers who are willing to pay more for higher-rated wines. Conversely, they might reconsider stocking wines like Rose, which have lower ratings and are among the lower-priced, to optimize their product offerings for better customer satisfaction and profitability.

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Conclusion

This analysis revealed that Sparkling wine has the highest average price and rating, while Rose wine has the lowest in both categories, with prices varying significantly across wine types. Gallo should focus on offering moderately priced, highly rated wines and target premium customers with Sparkling wine, while reconsidering the lower-rated Rose wine. Additionally, expanding their Sparkling wine offerings and adjusting inventory and marketing strategies can enhance profitability and customer satisfaction.

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Appendix

Figure 1- Red Wine Descriptive analysis

Rating			*Price*	
Mean	3.890338611		Mean	39.16511037
Standard Error	0.003314347		Standard Error	0.913610374
Median	3.9		Median	18.21
Mode	3.8		Mode	9.9
Standard Deviation	0.308305494		Standard Deviation	84.98541004
Sample Variance	0.095052278		Sample Variance	7222.519919
Kurtosis	0.003206425		Kurtosis	353.1198406
Skewness	-0.057264078		Skewness	13.40475759
Range	2.3		Range	3407.24
Minimum	2.5		Minimum	3.55
Maximum	4.8		Maximum	3410.79
Sum	33663.1		Sum	338895.7
Count	8653		Count	8653
Chris Williams				

Figure 2 - White Wine Descriptive Analysis

Rating			*Price*	
Mean	3.818700053		Mean	20.64039957
Standard Error	0.004325395		Standard Error	0.505211039
Median	3.8		Median	13.155
Mode	3.8		Mode	9.9
Standard Deviation	0.265016505		Standard Deviation	30.9542272
Sample Variance	0.070233748		Sample Variance	958.1641816
Kurtosis	0.327385784		Kurtosis	142.253876
Skewness	-0.047836478		Skewness	9.71633072
Range	2.2		Range	677.63
Minimum	2.7		Minimum	3.74
Maximum	4.9		Maximum	681.37
Sum	14335.4		Sum	77484.06
Count	3754		Count	3754
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Figure 3 - Sparkling Wine Descriptive Analysis

Rating			*Price*	
Mean	4.065942029		Mean	60.74271739
Standard Error	0.017156128		Standard Error	4.482181869
Median	4.1		Median	35.135
Mode	3.9		Mode	14.9
Standard Deviation	0.285019011		Standard Deviation	74.46359774
Sample Variance	0.081235837		Sample Variance	5544.827388
Kurtosis	-0.574942406		Kurtosis	8.88351574
Skewness	0.120145014		Skewness	2.701111008
Range	1.4		Range	488.5
Minimum	3.3		Minimum	6.5
Maximum	4.7		Maximum	495
Sum	1122.2		Sum	16764.99
Count	276		Count	276
Chris Williams				

Figure 4 - Rose Wine Descriptive Analysis

Rating			*Price*	
Mean	3.742710997		Mean	12.60363171
Standard Error	0.013846617		Standard Error	0.816335258
Median	3.8		Median	8.94
Mode	3.8		Mode	7.9
Standard Deviation	0.273799133		Standard Deviation	16.14198476
Sample Variance	0.074965965		Sample Variance	260.5636719
Kurtosis	0.70994998		Kurtosis	123.9274509
Skewness	0.087687377		Skewness	9.576956589
Range	2.1		Range	245.3
Minimum	2.7		Minimum	3.7
Maximum	4.8		Maximum	249
Sum	1463.4		Sum	4928.02
Count	391		Count	391
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Figure 5 Rating ANOVA Output

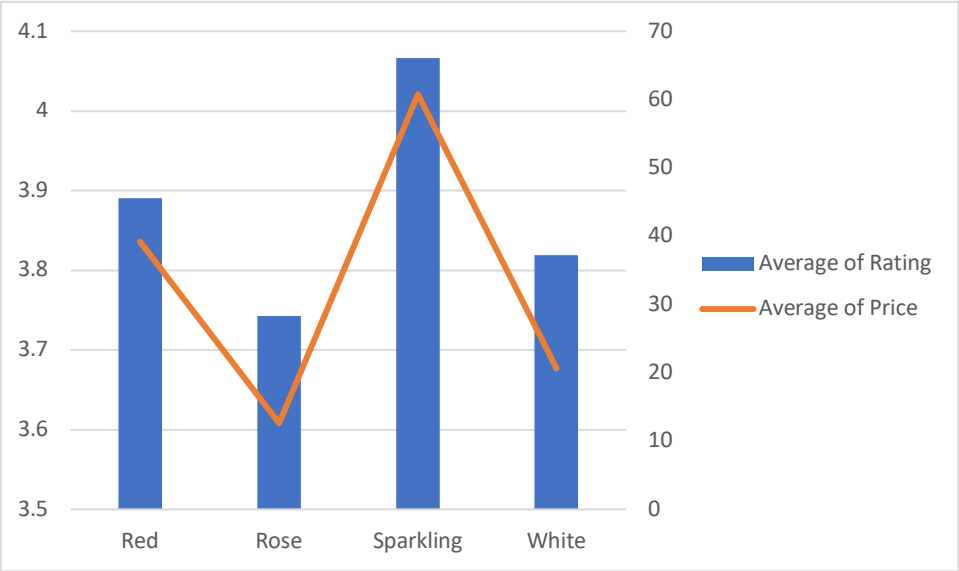
Anova: Single Factor						
SUMMARY						
Groups	*Count*	*Sum*	*Average*	*Variance*		
Red Rating	8653	33663.1	3.890338611	0.095052278		
White Rating	3754	14335.4	3.818700053	0.070233748		
Sparkling rating	276	1122.2	4.065942029	0.081235837		
Rose Rating	391	1463.4	3.742710997	0.074965965		
ANOVA						
Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	30.37892023	3	10.12630674	116.3466346	2.34555E-74	2.605588512
Within Groups	1137.556146	13070	0.087035665			
Total	1167.935066	13073				
Chris Williams						

Figure 6 - Price ANOVA Output

Anova: Single Factor						
SUMMARY						
Groups	*Count*	*Sum*	*Average*	*Variance*		
Red Price	8653	338895.7	39.1651104	7222.51992		
White Price	3754	77484.06	20.6403996	958.164182		
Sparkling Price	276	16764.99	60.7427174	5544.82739		
Rose Price	391	4928.02	12.6036317	260.563672		
ANOVA						
Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	1274071.69	3	424690.563	81.9756011	1.5451E-52	2.60558851
Within Groups	67711679.9	13070	5180.69471			
Total	68985751.6	13073				
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Figure 7 - Bar Graph by Average Rating and Price by Wine Type (Mac doesn't let you do Change Series Chart Type)



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